PAULO VITOR MENTANI

Brazilian, 26 years old
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Objective

Technical Product Specialist / Systems Integration Engineer

Education

• Industrial automation technologist, FATEC Osasco, 2019-2022 – finished

• Computer Engineering, UNIVESP, 2022-2027 – in curse

Work Experience

• *11/2024 – Atual – Vammo*

Job Title: Embedded System Engineer
- Development in C++ (using boost libraries, C++ 23, Qt Creator, QML) for embedded Linux, and creation of maintenance software for MS Windows.
- Development in C for embedded systems using ESP32 and STM32.
- Development in Python for embedded Linux using DBus, protobuf, alsa lib, and creation of metadata analysis scripts.
- Development in Python for Data Science and Machine Learning, using NumPy, Matplotlib, Pandas, and CUDA, applied to behavior analysis and failure prediction. -
Yocto development, package creation, BSP file.
- Development using the following protocols: Modbus RTU, RS485, CAN, I2C, MQTT, HTTPS, Bluetooth, LTE 4G.
- Hardware debugging, circuit analysis to identify root causes at the component level.
- Development of test jigs for the engineering team and test jigs for the operations team.
- Development of SOP (Supported Operating Procedure) for the operations team, field failure assessment, and creation of corrective strategies between engineering and operations teams.

• *06/2024 – 12/2024 – Pagmob*

Job Title: Firmware Developer
- Firmware development in C for embedded systems using FreeRTOS and ESP-IDF, along with embedded SQLite for local data storage.
- Firmware development for embedded Linux.
- Integration of payment systems using EMV for contactless card transactions in partnership with VISA. -
Integration of payment systems using BLE (Bluetooth Low Energy).
- Development of applications using HTTP and MQTT protocols.
- Implementation of the IEC 7816-4 standard with TLV-BER parsing in C.

• *04/2023 – 06/2024 – MAGNAMED*

Job Title: Software Developer
- Embedded Linux development, drivers, and image creation with Yocto. -
C++ programming using Qt Creator for Linux applications.
- Firmware development in C for Texas Instruments microcontrollers.
- Python scripting, utilizing libraries such as NumPy, Matplotlib, Pandas, and SciPy. -
Unit testing with GoogleTest and Parasoft, covering both unit and static tests.

- ***09/2022 – 04/2023 – Pasquali Solution (MAGNAMED)***

Job Title: C++ Developer

- C++ programming using Qt Creator for Linux applications.
- Firmware development in C for Texas Instruments microcontrollers.
- Python scripting, utilizing libraries such as NumPy, Matplotlib, Pandas, and SciPy. -
Unit testing with GoogleTest and Parasoft, covering both unit and static tests.

- ***10/2021 – 09/2022 – Power2Go***

Job Title: Firmware and Hardware Developer

- Firmware development in C and C++ for ESP32 and Atmega microcontrollers (using ESP-IDF, Arduino, FreeRTOS, and MQTT).
- Hardware development for energy consumption measurement and IoT communication. -
Creation of test devices for hardware and firmware testing.
- Creation of Python scripts using the AWS SDK (Boto3) and pySerial for applications on Windows and Linux

Qualification

- Experience in C, C++ , Python, MySQL e .NET;
- Linux development (Yocto Project, drivers' development).
- Unit test development using Google Test e Parasoft.

Languages

- English – Advanced
- Portugues - Native